

Arthur Krieger

Software/Platform Engineer · CS + Data Science

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SUMMARY

Software and platform engineer at Charles Schwab, on a team building the shared infrastructure other engineers build on. My main focus has been the metadata preservation system, which I designed and built end to end so warehouse governance metadata survives schema deployments, along with the Streamlit dashboards and dashboard template the data governance and tenant teams depend on. Outside work I design and operate KriegerDataForge, a multi-tenant data platform with its own OIDC identity provider, shared SDK, and Terraform control plane. I build systems that are idempotent, observable, and safe to re-run.

EXPERIENCE

Software/Platform Engineer — Charles Schwab

June 2025 - Present

Chicago, IL

- Architected and implemented a declarative metadata reconciliation system, mid April to end of August 2026, eliminating silent loss of warehouse governance metadata during schema deployments. Cut recovery of thousands of metadata rows from three to four hours of manual rework to seconds, with 20+ schema deployments absorbed since go-live and no human involvement in any of them
- Engineered it to be provably recoverable, with content hash change detection, fully resumable execution under a hard statement timeout, tiered retry with dead-lettering, and eight independent fail closed checks, so any failure mode is inaction rather than data corruption
- Made it object type agnostic and multi-tenant, so a new metadata category is a registry entry with zero engine changes and any team self-serves through a repo template, and built a spreadsheet to catalog generator so non-engineers author metadata with no YAML and no engineering dependency
- Took sole ownership of two business critical dashboard repositories after the original developers departed, rebuilt one on a feature based architecture, and published it as a template any tenant team can start from
- Optimized the dashboard's core Snowflake queries to roughly half the execution time with disk spill eliminated, serving 100+ daily users
- Build and maintain data pipelines delivering financial data to analysts and data scientists, develop GitHub Actions CI/CD workflows and DevOps automation, and provision GCP infrastructure with Terraform across five environments

Technologies. Python, Snowflake, SQL, GCP Cloud Services, GitHub Actions, Streamlit, Terraform, BitBucket, Bamboo

Data Engineer Intern — Charles Schwab

June 2024 - August 2024

Lone Tree, CO

- Part of a scrum team using Agile methodology. Built dashboards for data observability and monitoring of data pipelines using Streamlit and Snowflake
- Delivered Streamlit dashboards displaying pipeline operations and data profiling metrics using Snowflake
- Presented a research plan on using generative AI to synthesize data and improve fraud protection

Technologies. Python, Streamlit, Snowflake, SQL

Data Engineer Intern — Revantage (Blackstone Portfolio Co.)

June 2023 - August 2023

Chicago, IL

- Integrated Azure DevOps REST API in Databricks using Python and SQL to track software changes
- Researched and implemented External Tables in Snowflake to reduce costs and bypass unnecessary data transfers
- Presented to stakeholders at the end of each 3-week sprint

Technologies. Python, SQL, Databricks, Azure

Research Assistant — Department of Obstetrics and Gynecology — Wayne State University

September 2021 - August 2022

Detroit, MI

- Used R to analyze gene expression rates and auto generate visualizations for a published preeclampsia research paper
- Built an R Shiny app for researchers to query an SQL database and generate plots interactively

Technologies. R, SQL, ggplot2, RShiny

PROJECTS

KriegerDataForge — Multi-Tenant Platform Ecosystem

A multi-tenant data platform spanning 18 repositories, built and operated solo. An OAuth 2.0 and OIDC identity provider issuing RS256 JWTs with a JWKS endpoint for cross-app SSO, a shared Python SDK every tenant backend installs instead of reimplementing auth, a Terraform control plane declaring every environment, and a centralized CI/CD library inherited by every repo.

Technologies. Next.js, React, TypeScript, TailwindCSS, FastAPI, Python, SQLAlchemy, PostgreSQL, Alembic, OAuth 2.0 / OIDC, Terraform, GitHub Actions, Docker, Vercel, GCP

Tiffany's Space — E-commerce Platform · KriegerDataForge

A production boutique e-commerce storefront. Server authoritative on every price, total, and inventory count, with a two step reserve then pay checkout that is transactional, idempotent, and concurrency safe, plus an admin dashboard for catalog and order management.

Technologies. Next.js, React, TypeScript, TailwindCSS, FastAPI, PostgreSQL, SQLAlchemy, Stripe, Zod

Calorie Tracker — Nutrition App · KriegerDataForge

A free forever nutrition tracker on a 41 field food database, with custom foods and meals, a guided creation wizard, and all nutrition totals recomputed server side so client submitted values are never trusted.

Technologies. Next.js, React, TypeScript, FastAPI, Python, PostgreSQL, SQLAlchemy

EDUCATION

University of Michigan College of Engineering

Sep 2022 - May 2025

B.S. Computer Science Engineering, Data Science Minor — GPA 3.75 / 4.0 · Ann Arbor, MI

Coursework. EECS 281 - Data Structures & Algorithms, EECS 370 - Computer Architecture, EECS 442 - Computer Vision, EECS 445 - Intro to ML, EECS 449 - Conversational AI, EECS 481 - Software Development, EECS 485 - Web Systems, EECS 492 - Intro to AI, DATASCI 315 - Deep Learning

Michigan State University College of Engineering

Sep 2021 - May 2022

B.S. Computer Science (transferred), Honors College — GPA 3.93 / 4.0 · East Lansing, MI

SKILLS

Languages. Python, TypeScript, JavaScript, C/C++, SQL, R, HTML/CSS

Backend. FastAPI, REST APIs, SQLAlchemy, SQLAlchemy, Pydantic, Alembic

Data & Cloud. Snowflake, Databricks, PostgreSQL, GCP, Azure, AWS, Vercel, Streamlit, Pandas, PyTorch, TensorFlow

Testing & Quality. pytest, Playwright, Ruff, mypy, ESLint, CodeQL

Frontend. Next.js, React, Tailwind CSS, Zod, Jinja

Auth & Security. OAuth 2.0, OIDC, JWT / JWKS, PKCE, argon2id, OWASP, Bandit

DevOps & Infrastructure. Terraform, GitHub Actions, Docker, CI/CD, Bitbucket, Bamboo

Tools. Git / GitHub, Jira, Confluence, Agile / Scrum, Power BI, Excel

CERTIFICATIONS

• Snowflake SnowPro Core - October 2025

• Microsoft Certified: Azure Fundamentals - August 2023

• Google Cloud Certified: Associate Cloud Engineer - August 2025

• Academy Accreditation: Databricks Lakehouse Fundamentals - August 2023

SPEAKING

Guest Lecturer, EECS 481 Software Engineering

November 2025

University of Michigan, College of Engineering · Prof. Westley Weimer

Invited by Professor Westley Weimer to return to the University of Michigan and deliver one of four annual guest lectures, selected from a pool of alumni within six months of graduating. Presented to 100+ students on the path from university to industry, which coursework translates most directly to industry skills, and how to secure internships, followed by an extended Q&A. Also joined Professor Weimer's lunch with Master's and PhD students to discuss research and industry practice.

Lecture. youtu.be/3r0xsfJqhnw · **Q&A.** youtu.be/bcbiZgvfEPM